



Mae Fah Luang University

School of Applied Digital Technology



Computer Engineering

**** รับสมัครนักศึกษาปริญญาเอก ประจำปีภาคการศึกษา 1/2565 ****

Programme

Degree Academic Programme Plan Class Length of Programme

Ph.D.	Computer Engineering	1.1, 2.1	Mon-Fri	3 years
		2.2	Mon-Fri	5 years

Programme's Philosophy

Computer engineering is a discipline related to the design and construction of computers or computer systems and systems that use computers. It is directly related to hardware, software, and communication. It plays a crucial role in industrial reform for economic and social development from the past to the present and is also essential for dealing with various changes that are happening in the present and future. Since the Third Industrial Revolution in 1969 until now, disruptive technologies have caused many things to change from what they used to be, impacting businesses and people's lifestyles.

However, not all newly emerging technologies can completely change the nature of business or society. Only certain types of technology have true potential to significantly change the current situation. Particularly, computer engineering remains crucial for inventing new theories and engineering innovations to deal with these changes urgently and continuously through the development of human resources in this field.



The important foundation of this curriculum is to integrate Constructivist Theory with Personalized Learning. Constructivist Theory combines cognitive constructivism and social constructivism, believing that learners construct knowledge by linking experiences or new information received from the environment with existing knowledge to create their understanding. They use problem situations or intellectual conflicts arising from social interactions with others to create new intellectual structures. Therefore, the curriculum utilizes teaching and learning management processes through problem or project-based learning to support these theories, focusing on allowing learners to face real situations, search for solutions independently with the support of teachers, and interact with others.

Personalized Learning involves allowing learners to participate in determining the direction of learning according to their interests or goals, based on the belief that learners differ in preferences, talents, and abilities. Each learner should have the opportunity to set learning goals and choose what they want to learn independently. The curriculum, therefore, manages teaching and learning based on learners' competencies, using flexible time and methods according to the nature of each learner to develop learners efficiently throughout the education process, with learning support activities, strong advisory teams, and aligned competency assessment systems.

This curriculum aims to produce graduates who can create commercial knowledge and engineering innovations in computer engineering to solve real-world problems and meet the usage needs of diverse target groups. Learners have the freedom to define research topics and plan learning paths independently, with advisory teams providing guidance, monitoring, assessment, and educational support activities throughout the curriculum.

Programme's Objectives

To produce graduates with the following knowledge and abilities:

1. Advanced research knowledge and skills in computer engineering, communication, and information.
2. Strong commitment to professional ethics.
3. Ability to develop new knowledge and technical innovations for commercial advancement.



4. Collaboration skills to work with academic networks, research, and innovation.

5. Capability to disseminate new knowledge or original innovations at local, national, regional, and international levels.

Programme's Highlights

The curriculum focuses on creating fundamental knowledge in artificial intelligence, communication systems, processing, and automation platforms through research processes that can demonstrate prototypes or formats for industrial or commercial application.

The curriculum enhances doctoral graduates to act with a focus on success in advanced computer engineering research development.

The curriculum manages teaching and learning using the foundation of Constructivist Theory integrated with Personalized Learning, emphasizing learners facing real situations, independently searching for problem-solving strategies with the support of teachers and interaction with others.

Career Opportunities

- Computer engineering, communication, and information technology academic.
- Computer engineering, communication, and information technology researcher and innovator.
- Computer engineering, communication, and information technology engineer and consultant.
- Computer engineering, communication, and information technology entrepreneur.

Program Learning Outcomes: PLOs

- PLO1: Create fundamental knowledge in artificial intelligence, communication systems, processing, and automation platforms through research processes.



- Sub-PLOs 1.1: Analyze and link key knowledge of disruptive technologies in artificial intelligence, communication systems, processing, and automation platforms.
- Sub-PLOs 1.2: Generate new knowledge by connecting knowledge in artificial intelligence, communication systems, processing, and automation platforms through research processes.
- PLO2: Develop technical innovations or advance fundamental knowledge for societal or commercial benefits.
 - Sub-PLOs 2.1: Present advancements in knowledge in artificial intelligence, communication systems, processing, and automation platforms.
 - Sub-PLOs 2.2: Develop and transfer technical innovations in artificial intelligence, communication systems, processing, and automation platforms for societal or commercial benefits at local, national, regional, and international levels.
- PLO3: Practice general research ethics and human research ethics to create new knowledge.
 - Sub-PLOs 3.1: Apply human research ethics in research related to humans.
 - Sub-PLOs 3.2: Conduct research processes, data collection, experiments, and straightforward result analysis.
 - Sub-PLOs 3.3: Produce academic works with honesty, without plagiarizing others' works.
- PLO4: Demonstrate self-directed learning, continuous self-improvement, and adaptability in teamwork ready for changes.
 - Sub-PLOs 4.1: Analyze academically with confidence, listen to different opinions, and adapt to teamwork.
 - Sub-PLOs 4.2: Choose self-directed learning paths and continuously self-improve.
- PLO5: Act with a focus on success in advanced computer engineering development tasks.
 - Sub-PLOs 5.1: Set academic success goals related to advanced computer engineering.
 - Sub-PLOs 5.2: Act according to success goals.

Tuition and Fees

- **Plan 1.1, 2.1:**
 - 58,500 Baht per semester
 - 351,000 Baht per programme



- **Plan 2.2:**
 - 58,500 Baht per semester
 - 585,000 Baht per programme

Programme Structure

Plan 1.1

Total credits : 48 credits (3-year Doctor's Degree)

1. Dissertation 48 credits

Plan 2.1

Total credits : 48 credits (3-year Doctor's Degree)

1. Major Required Courses 9 credits
 2. Major Elective Courses 3 credits
 3. Dissertation 36 credits

Plan 2.2

Total credits : 72 credits (5-year Doctor's Degree)

1. Major Required Courses 18 credits
 2. Major Elective Courses 6 credits
 3. Dissertation 48 credits

(Refer to Doctor of Philosophy Program in Computer Engineering Year 2023)

Links:

Plan 1.1

Plan 2.1

Plan 2.2

Note:

- **1.1** = Dissertation only (not less than 48 credits) (Master's degree graduates)
- **1.2** = Dissertation only (not less than 72 credits) (Bachelor's degree with honours graduates)
- **2.1** = Course work (not less than 12 credits) with Dissertation (not less than 36 credits) (Master's degree graduates)
- **2.2** = Course work (not less than 24 credits) with Dissertation (not less than 48 credits) (Bachelor's degree with honours graduates)



รายละเอียดหลักสูตรเพิ่มเติม

More details (in Thai) : <<

[link](#)

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More details (in English) : <<

[link](#)

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Doctoral Degree

Computer Engineering



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Mae Fah Luang University

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